

4. Central Operation – placeOrder

Operation: placeOrder

Purpose: To place a food order for a user using the selected cart items and complete the required payment.

4.1 Inputs

Input	Description
User information	Information needed to identify and verify the user placing the order.
Cart information	The selected food items and their quantities.
Payment information	The payment method and information needed to complete payment.

4.2 Outputs

Output	Description
Order confirmation	Confirmation that the order was successfully placed.
Order status	The current status of the new order.
Payment status	The result of the payment operation.

4.3 Error Cases

Error Case	Meaning
User not verified	The user cannot be verified.
Cart is empty	There are no items to order.
Item unavailable	One or more selected items are unavailable.
Payment failed	The payment could not be completed.
Order cannot be placed	The order could not be completed because of a processing or service failure.

4.4 Internal Details Hidden from Callers

Hidden detail	How it works internally
Order storage	The service keeps the order information internally and manages its order status without requiring the caller to know how the information is stored.
Cart processing	The service reads the selected items and quantities from the cart, checks that the cart can be converted into an order, and uses the valid items for the order.
Amount calculation	The service calculates the total amount from the selected items and quantities, including any applicable charges defined by the system.
Item availability	The service checks with the Catalogue Service to make sure the selected items are available before confirming the order.
Payment processing	The service sends the required payment request to the Payment Service and uses the returned payment result to decide whether the order can continue.
Order status update	After successful processing, the service changes the order to the appropriate status and returns that status to the caller.

Hidden detail	How it works internally
Communication with other services	The caller does not need to know how Cart & Order communicates with User & Auth, Catalogue, or Payment. These service calls are handled internally.
Database implementation	The caller does not see database tables, SQL queries, primary keys, foreign keys, or ID formats. These are implementation details hidden inside the service.

4.5 Functionality of placeOrder

Step 1 – Verify user: The operation first makes sure that the user is valid and allowed to place an order.

Step 2 – Validate cart: It checks that the cart contains items and that the selected quantities are valid.

Step 3 – Check availability: It asks the Catalogue Service whether the selected items are currently available.

Step 4 – Calculate amount: It determines the amount that needs to be paid for the selected order.

Step 5 – Process payment: It requests payment through the Payment Service. If payment fails, the order is not confirmed.

Step 6 – Confirm order: After successful payment and validation, the service confirms the order and updates the required status.

Step 7 – Return result: It returns the order confirmation, order status, and payment status to the caller.